

ASSIGNMENT - 2 :

Encrypt/Decrypt using Pen and Paper method:

1. Encrypt "CHASE YOUR DREAMS" using

(a) Caesar Cipher:

As we know, Caesar cipher uses shift, $n=3$

$$C = (P + 3) \bmod 26.$$

CHASE $\Rightarrow$ C H A S E

5 10 3 21 7

5 10 3 21 7

F K D V H

YOUR $\Rightarrow$ Y O U R

27 17 23 20

1 17 23 20

B R X U

DREAMS $\Rightarrow$ D R E A M S

6 20 7 3 15 21

6 20 7 3 15 21

G U H D P V

Cipher Text : F K D V H B R X U G U H D P V

(b) Shift cipher with Key = 7: $C = (P + 7) \bmod 26$

CHASE $\Rightarrow$ C H A S E

9 14 7 25 11

9 14 7 25 11

$\Rightarrow$ J D H Z L

YOUR $\Rightarrow$ Y O U R

31 21 27 24

5 21 1 24

F V B Y

DREAMS $\Rightarrow$ D R E A M S

10 24 11 7 19 25

10 24 11 7 19 25

K Y L H T Z.

Cipher Text: JOHZL FVB Y KYLHTZ.

(c) ROT 13:

we shift = 13 positions in alphabet.

Ciphertext = $(P+13) \bmod 26$.CHASE $\Rightarrow$ C H A S E

15 20 13 31 17

15 20 13 5 17

P U N F R

YOUR $\Rightarrow$ Y O U R

37 27 33 30

11 1 7 4

L B H E

DREAMS $\Rightarrow$ D R E A M S

16 30 17 13 25 31

16 4 17 13 25 5

Q E R N Z F

Cipher Text: PUNFR LBHE QERNZF

2. Decrypt:

(a) "WRWDOOB VHFUHW PHVVVDJH"

using ceaser cipher:

Plain text: $(C-3) \bmod 26$.

WRWDOOB $\Rightarrow$ W R W D O O B

22 17 22 3 14 14 1

19 14 19 0 11 11 -2

19 14 19 0 11 11 24

T O T A L L Y

VHFUHW $\Rightarrow$ V H F U H W

21 7 5 20 7 22

18 4 2 17 4 19

18 4 2 17 4 19

S E C R E T

PHVVVDJH $\Rightarrow$ P H V V D J H

15 7 21 21 3 9 7

12 4 18 18 0 6 4

12 4 18 18 0 6 4

M E S S A G E

Plain text: TOTALLY SECRET MESSAGE

(b) (i) "MJQQT BTWQI" using shift cipher = 5.

Plain Text = $(C - 5) \bmod 26$

MJQQT $\Rightarrow$ M J Q Q T
 12 9 16 16 19
 7 4 11 11 14
 7 4 11 11 14
 H E L L O

BTWQI $\Rightarrow$ B T W Q I
 1 19 22 16 8

4 14 17 11 3
 22 14 17 11 3
 W E O R L D

Plaintext: HELLO WORLD.

(ii) STB DTWP YNRJ using shift cipher

Plaintext = $(C - 5) \bmod 26$

STB $\Rightarrow$ S T B
 18 9 1
 13 4 -4
 13 4 22
 N E W

DTWP $\Rightarrow$

D	T	W	P
3	19	22	15
-2	14	17	10
24	14	17	10
Y	O	R	K

YNRJ $\Rightarrow$

Y	N	R	J
24	13	17	9
19	8	12	4
19	8	12	4
T	I	M	E

Plaintext : New YORK TIME

(c) " FRAQ ZBER PBSSER " using ROT 13

Plaintext = $(C - 13) \bmod 26$

FRAQ $\Rightarrow$

F	R	A	Q
5	17	0	16
-8	4	-13	3
18	4	13	3
S	E	N	D.

ZBER $\Rightarrow$

Z	B	E	R
25	1	4	17
12	-12	-9	4
12	14	17	4
M	O	R	E

PBSSRR $\Rightarrow$ P B S S R R

15	1	18	18	17	17
2	-12	5	5	4	4
2	14	5	5	4	4
C	O	F	F	E	E

Plaintext : SEND MORE COFFEE